ESG 633 Programming for Geoinformatics

Lab 4: Satellite Data Processing

Download a Sentinel-2, Landsat 8/9 small patch area of 10km^2 of your hometown/favorite place and do the following:

1. Use rasterio to open the GeoTIFF file. Print out the dimensions of the satellite image. Find the total bands available in the data and Coordinate Reference System (CRS)
2. Read this TIFF file and compute its minimum, maximum, mean across bands. Use matplotlib to plot a histogram of 3 selected spectral bands (3 histograms seperately).
3. Do bandstacking by stacking all the bands of the satellite image and save it as a Geotiff file.
4. Create and display True Color Composite (Natural Color, e.g., RGB mapped to Red, Green, Blue bands) and False Color Composite (Color Infrared, e.g., RGB mapped to NIR, Red, Green bands).
5. Clip a small area of interest from the satellite image and save it as a separate file.
6. Plot the spectral signature curve for different Land use categories (band wavelength vs. reflectance). For example: Spectral plot for pixels with the following landuse categories : Vegetation, water, urban, builtup.
7. Compute the NDVI of the image and using band thresholding (on NDVI values) compute the area of vegetated land in the image. Create a map of the vegetated landcover in a map layout (Plot the map with band values containing 0-No vegetation, 1-Vegetation).